Friis Transmission Formula

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Lecture Objective

By the end of this lecture, you will be able to:

- Evaluate the efficacy of a communications link using the Friis transmission formula.

Friis Transmission Formula (1 of 2)

- Relates the power received by one antenna to the power transmitted by another.
- Assumes each antenna is in the far-field of the other.
- Assumes free space propagation.

Friis Transmission Formula (2 of 2)

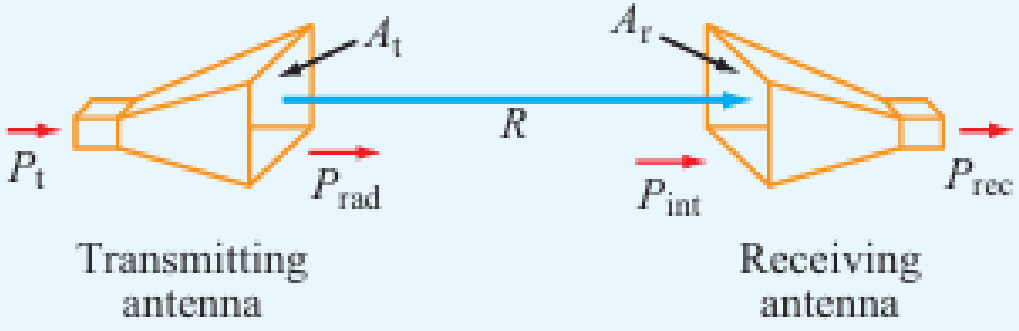

Figure 9-19 Transmitter–receiver configuration.

Derivation of Friis Formula (1 of 3)

- If the transmit antenna were isotropic, the power density at the receive antenna would be

$$S_{\text{iso}} = \frac{P_t}{4\pi R^2}$$

- Since the transmit antenna is not isotropic, the actual power density at the receive antenna is given by

$$S_{\text{r}} = \frac{P_t G_t}{4\pi R^2}$$

realized gain of transmit antenna in direction of receiver

Question

- The factor $G_t P_t$ is called *effective isotropic radiated power* (EIRP). This factor shows how antenna gain can be utilized to increase the signal at the receiver. For example, if the transmitter power is 100 watts and the antenna gain is 20 dBi (dB relative to isotropic antenna), the EIRP is 10,000 watts. However, why is it that a high gain antenna is not suitable for all applications?

Derivation of Friis Formula (2 of 3)

- The receive antenna intercepts some of the transmitted power through its effective area.

$$P_{int} = \frac{P_t G_t}{4\pi R^2} A_r$$

- Recall that

$$A_{\mathrm{r}} = \frac{\lambda^2 D_{\mathrm{r}}}{4\pi}$$

realized gain of receive antenna in direction of transmitter

$$P_{rec} = \xi_r P_{int}$$

Derivation of Friis Formula (3 of 3)

- Thus,

$$P_{\text{rec}} = \xi_{\text{r}} P_{\text{int}} = \frac{P_{\text{t}} G_{\text{t}}}{4\pi R^2} \xi_{\text{r}} A_{\text{r}} = \frac{P_{\text{t}} G_{\text{t}}}{4\pi R^2} \frac{\lambda^2 G_{\text{r}}}{4\pi}$$

- Finally, we have the most useful and commonly presented form of the equation

$$\boxed{\frac{P_{\text{rec}}}{P_{\text{t}}} = G_{\text{t}} G_{\text{r}} \left(\frac{\lambda}{4\pi R} \right)^2}$$

Path Loss Factor

- The free space path loss factor is $\left(\frac{\lambda}{4\pi R}\right)^2$
- This factor results from the spherical spreading of the power from the transmit antenna in free space.
- Propagation in other environments (e.g., in urban terrain, inside buildings, etc.) can be modeled by using more sophisticated – even statistical – path loss factors.

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